

WELL_NECT

CareNexus — Professional 4-Week Development Roadmap

Project: Intelligent Hospital Service Comparison and Navigation Platform

Core principle: Required Service → Confirm Availability → Find Suitable Hospitals → Compare/Rank → Select Hospital → Route & Navigation.

1. Project Development Strategy

This roadmap keeps the completed Week 1 Member 1 and Member 2 work unchanged and continues with the dependency-based CareNexus development model. Each member completes and validates the required output before the dependent member proceeds.

Member	Week 1 Responsibility	Primary Output	Status
Member 1	Hospital Data Model	Hospital schema and sample data	Completed
Member 2	Service Catalog	Standardized medical services	Completed
Member 3	Service Availability Mapping	Hospital-service availability mapping	In Progress
Member 4	Database Integration	Unified PostgreSQL database	Pending

2. Week 1 — Hospital & Service Foundation

Member 1 — Hospital Data Model: Hospital ID, name, address, city, latitude, longitude, contact and rating.

Member 2 — Service Catalog: Emergency, ICU, MRI, CT Scan, Cardiology, Orthopedics, Blood Bank and Pharmacy, with descriptions and categories.

Member 3 — Service Availability Mapping: Connect each hospital to medical services using the statuses **Available, Limited, Unavailable, and Unknown**. The project currently uses demo/seed availability data; it should not be presented as real-time hospital availability.

Member 4 — Database Integration: Ensure hospitals, services and availability mappings are integrated and validated in PostgreSQL.

Week 1 milestone: The system can answer whether a hospital provides a requested medical service.

3. Week 2 — Backend Intelligence

Order	Member	Feature	Expected Handover
1	Member 1	FastAPI Foundation	Running API, database connection and base endpoints
2	Member 2	Hospital Search	Search by hospital name, city/location and stored fields
3	Member 3	Service Availability Filter	Only hospitals matching requested services
4	Member 4	Comparison Engine	Comparison of 2–4 suitable hospitals

Week 2 milestone: A backend request such as “MRI + Emergency” returns hospitals that satisfy the requested service requirements and allows suitable hospitals to be compared.

4. Week 3 — Location, Recommendation & Route

Week 3 adds location intelligence and navigation. The key rule is that proximity must never override service availability: a hospital should only enter the recommendation stage after satisfying the requested service requirements.

Member	Responsibility	Implementation
Member 1	Location Foundation	Obtain user latitude/longitude and use hospital coordinates from the database.
Member 2	Nearby Hospital Logic	Calculate distance only for service-matched hospitals.
Member 3	Smart Ranking	Rank suitable hospitals using service match, distance, rating and other available criteria
Member 4	Route & Navigation	Integrate a maps/routing provider after the user selects a recommended hospital.

Recommended Ranking Model

For a stronger recommendation system, use service suitability as the highest-priority factor, followed by distance and other stored criteria. A possible project scoring model is: Service Match 50%, Distance 25%, Hospital Rating 15%, Availability 10%. This is a proposed enhancement to the roadmap, not a requirement stated in the original PDF.

Week 3 flow

Get Location → Find Service-Matched Hospitals → Calculate Distance → Compare/Rank → Select Hospital → Generate Route

5. Week 4 — Frontend, Integration & Testing

Member	Responsibility	Expected Output
Member 1	Search Interface	Service selection, hospital search and location input UI
Member 2	Hospital Results UI	Service-matched hospital cards/results
Member 3	Comparison & Recommendation UI	Side-by-side comparison and ranked recommendations
Member 4	Map, API Integration & Testing	Connected APIs, maps/routes, loading/error states and end-to-end testing

6. Data & Real-Time Enhancement Strategy

The current Week 1 availability records are demo/seed data. The roadmap does not specify a particular live hospital-data provider. Therefore, the project should not claim that the current records are real-time. A future enhancement can introduce a data-update/integration layer that receives verified updates from an appropriate hospital or external data source and stores the updated availability in PostgreSQL.

Recommended architecture: External/Verified Hospital Data Source → CareNexus Data Update Layer → PostgreSQL → Availability Verification → Filtering → Distance → Ranking → Recommendation → Navigation.

7. Final System Architecture

Frontend: React + JavaScript/TypeScript + CSS/Tailwind CSS

Backend: Python + FastAPI

Database: PostgreSQL + SQLAlchemy

Maps: Maps/Routing API

Version Control: Git + GitHub

Testing: Pytest + frontend testing tools

8. Final CareNexus User Flow

1	User selects required medical service
2	CareNexus verifies service availability
3	System identifies suitable hospitals
4	User location is obtained
5	Distance is calculated for suitable hospitals
6	Hospitals are compared and ranked
7	Best matching hospital is recommended
8	User selects a hospital
9	Route and navigation are generated

Project Title: CareNexus: An Intelligent Hospital Service Comparison and Navigation Platform

Tagline: Find the Right Service. Compare the Right Hospitals. Choose the Right Route.